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Numerical Solutions of frequency dependence of real and imaginary parts of the complex conductivity in Eliashberg theory

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Introduction

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Chapter 1

BCS theory

...BCS SHORT REVIEW

1.1 Mattis Bardeen formula

The Mattis-Bardeen formula (MB) describes the complex conductuvity $\sigma(\omega, T)$ of superconductors in the BCS theory framework, so it is accurate for weak-coupling superconductors materials. The originally form was derived by Mattis and Bardeen in 1958, and also assumes the clean limit $l \to \infty$ and the Pippard limit $\xi \leq \lambda_L$. Then Nam showed in 1967 that the formula is valid also in the dirty limit $l \to 0$ and the local limit $\xi \gg \lambda_L$.
The MB equations are given by the two integrals

$$\begin{aligned}\frac{\sigma_1(\omega,T)}{\sigma_n} = \frac{2}{\hbar\omega}\int_{\Delta}^{\infty} \frac{[f(E)-f(E+\hbar\omega)]\left[E^2+\Delta^2+\hbar\omega E\right]}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE \\ + \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega}^{-\Delta} \frac{[1-2f(E+\hbar\omega)]\left[E^2+\Delta^2+\hbar\omega E\right]}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE,\end{aligned} \tag{1.1}$$

for the real part and by

$$\frac{\sigma_2(\omega,T)}{\sigma_n} = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega,\,-\Delta}^{\Delta} \frac{[1-2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{\Delta^2-E^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE, \tag{1.2}$$

for the imagimary part, where the σ_n is the normal state conductivity and $f(E) = \frac{1}{1+exp(\frac{E}{TK_B})}$ is the Fermi-Dirac distribution. The lower limit in the integral of Eq. 1.2 is $-\Delta$ if $\hbar\omega > 2\Delta$. The expression for the real part of the conductivity describes the response of the normal carriers, that is, the dissipative part. The first term of Eq. 1.1 represents the effects of thermally excited quasiparticles, which vanishes when $T = 0$. The second term mainly accounts for the contribution of photon-excited quasiparticles; it requires the breaking of a Cooper pair, so it is basically zero for $\hbar\omega < 2\Delta(T)$. How- ever, there is a small contribution possible of quasiparticles thermally excited across the gap under the critical temperature.

1.1.1 Numerical Integration of the Mattis-Bardeen formula

Clearly Eqs.1.2 and 1.1 are not analytically integrable, so we need to use numerical methods to evaluate them. In this work i used the standard integrating methods in MATLAB, being careful with the singularities in the limits of integration. For the sake of simplicity of notation let's define the following integrals from the MB equations:

$$I_1 = \frac{2}{\hbar\omega}\int_{\Delta}^{\infty} \frac{[f(E)-f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE, \tag{1.3}$$

$$I_2 = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega}^{-\Delta} \frac{[1-2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{E^2-\Delta^2}\sqrt{(E+\hbar\omega)^2-\Delta^2}}\,dE \quad \text{and} \tag{1.4}$$

$$I_3 = \frac{1}{\hbar\omega}\int_{\Delta-\hbar\omega,-\Delta}^{\Delta} \frac{[1-2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{\Delta^2-E^2}\sqrt{\Delta^2-(E+\hbar\omega)^2}}\,dE. \tag{1.5}$$

The problems are in the square root terms in the denominators, which lead to singularities. To overcome this issue, i used a change of variable in the integrals that cancels out the singular terms in the denominator.

For the integral I_1 we have a problem in the lower limit $E = \Delta$, but using

$$E = \Delta\ cosh(u) \quad \leftrightarrow \quad u = cosh^{-1}(\frac{E}{\Delta}),$$

so that

$$du = \frac{dE}{\sqrt{E^2-\Delta^2}} \quad \Rightarrow \quad dE = \sqrt{E^2-\Delta^2}du$$

and the integral becames

$$I_1 = \frac{2}{\hbar\omega}\int_{0}^{\infty} \frac{[f(\Delta cosh(u))-f(\Delta\ cosh(u)+\hbar\omega)]\left[\Delta^2(cosh^2(u)+1)+\hbar\omega\Delta\ cosh(u)\right]}{\sqrt{(\Delta\ cosh(u)+\hbar\omega)^2-\Delta^2}}\,du. \tag{1.6}$$

Now we can integrate numerically with the standard methods in MATLAB sush as .

The integrals I_2 and I_3 have singularities both in the lower and upper limits, we can write them as

$$\frac{1}{\hbar\omega}\int_{a}^{b} \frac{[1-2f(E+\hbar\omega)]\left(E^2+\Delta^2+\hbar\omega E\right)}{\sqrt{b^2-E^2}\sqrt{(E^2-a^2}}\,dE. \tag{1.7}$$

Then with the change of variables

$$E = \frac{b-a}{2}\ sin(t) + \frac{b+a}{2} \quad \leftrightarrow \quad t = sin^{-1}\left(\frac{2E-(b+a)}{b-a}\right),$$

that leads to

$$dt = \frac{dE}{\sqrt{b-E}\sqrt{E-a}} \Rightarrow dE = \sqrt{(b-E)(E-a)}dt,$$

we can rewrite Eq. 1.7 as

$$\frac{1}{\hbar\omega}\int_{-\pi/2}^{\pi/2}\frac{\left[1-2f\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}+\hbar\omega\right)\right]\left[\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}\right)^2+\Delta^2+\hbar\omega\left(\frac{b-a}{2}\ sin(t)+\frac{b+a}{2}\right)\right]}{\sqrt{(b+E)(E+a)}}\,dt. \tag{1.8}$$

And the singularities have been removed.

1.2 Zimmermann formula for finite scattering rate

The Zimmerman forrmula is a generalization of the Mattis Bardeen equations for finte scattering time τ, this is parametrized by the scattering rate $\gamma = 1/\tau$. This parameter depends both on the material and the temperature. While the Eqs. 1.1 and 1.2 are just valid for the two limit cases $\tau \to 0$ and $\tau \to \infty$ (dirty and clean limit respectively), the following equations are valid for arbitrary elastic scattering rate of the carriers in the material.

1.2.1 Numerical integration of Zimmermann formula

Chapter 2

Results with BCS theory

Now that we can solve Eqs. 1.1 and 1.2 let's see the results for some different materials and various temperatures. In particular i have tested Nb, Pb and Al.

For different materials we need the experimental values of thecritical temperature T_c, under which they become superconductors, and the zero temperature value of the enrgy gap Δ_0. Actually for Δ_0 we extract the value from the BCS relation

$$\frac{2\Delta_0}{K_B T_c} = \alpha, \tag{2.1}$$

where α is ~ 3.53 for weak-coupling materials, that is the limit where BCS theory and MB formulas holds, and is greater for strong-coupling materials such as Nb and Pb. In Tab. 2.1 below the T_c and α values are showed.

Since $\Delta = \Delta(T)$ is a function of the temperature and is zero at $T \geq T_c$ i have extrapolate it in the BCS approximation as

$$\Delta = \Delta_0 \ \tanh\left(e^{\gamma}\sqrt{\frac{8}{7\xi(3)}}\ \sqrt{\frac{T_c}{T}-1}\right) \simeq \Delta_0 \ \tanh\left(1.74\ \sqrt{\frac{T_c}{T}-1}\right). \tag{2.2}$$

In the following we see some test for Nb, Pb and Al at different temperatures. In Fig. 2.1 i compared the real and imaginary conductivity for Nb, Pb and Al at $T = 0.8T_c$. In Figs.2.4, 2.2 and 2.3 we see conductivity for different temperatures in the various materials. The last Fig. 2.5 i have plotted the conductivity of Niobium with the energy scale normalized by $2\Delta_0$ to check if the real part of σ/σ_n correctly drops down at $\hbar\omega = 2\Delta_0$ when $T = 0$, and if increasing T the gap moves to lower energies/frequencies.

Table 2.1: Superconducting materials experimental parameters.

Material	T_c(K)	α
Nb	9.3	3.65
Pb	7.2	3.95
Al	1.2	3.53

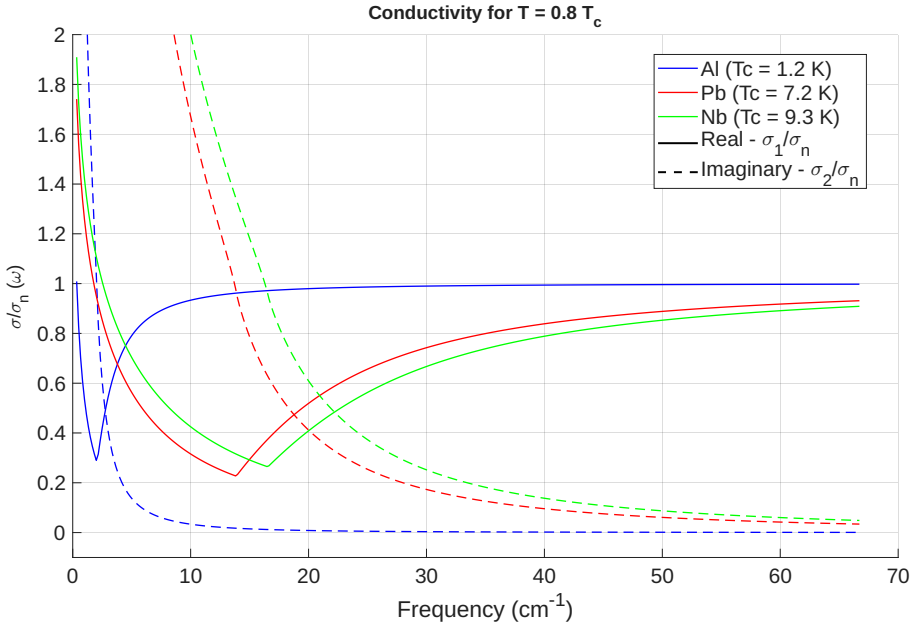


Figure 2.1: Complex conductivity for different materials.

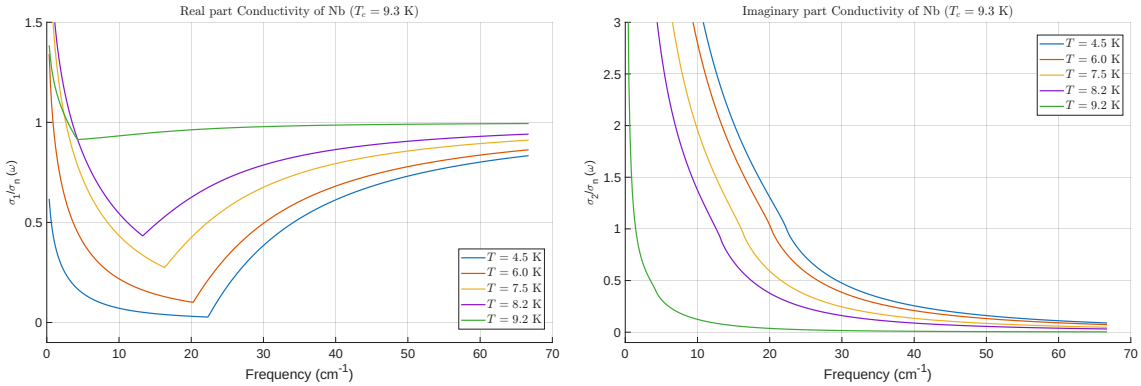


Figure 2.2: Real (sx) and imaginary (dx) conductivity for Nb.

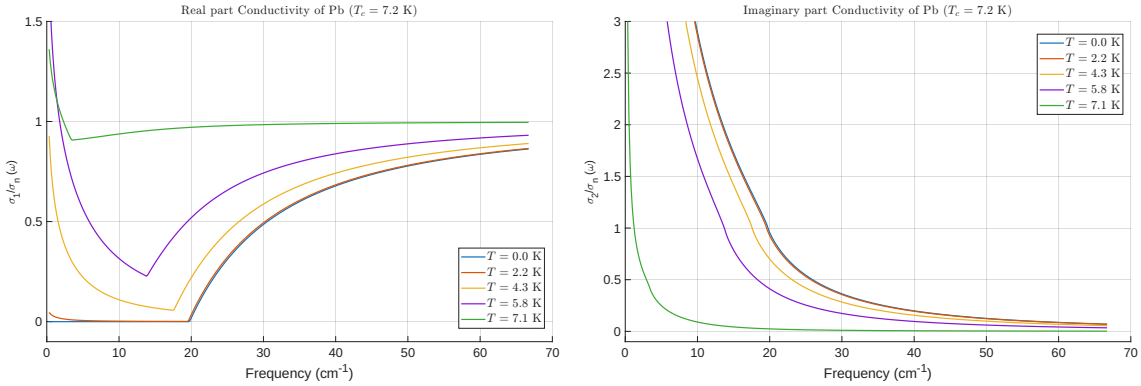


Figure 2.3: Real (sx) and imaginary (dx) conductivity for Pb.

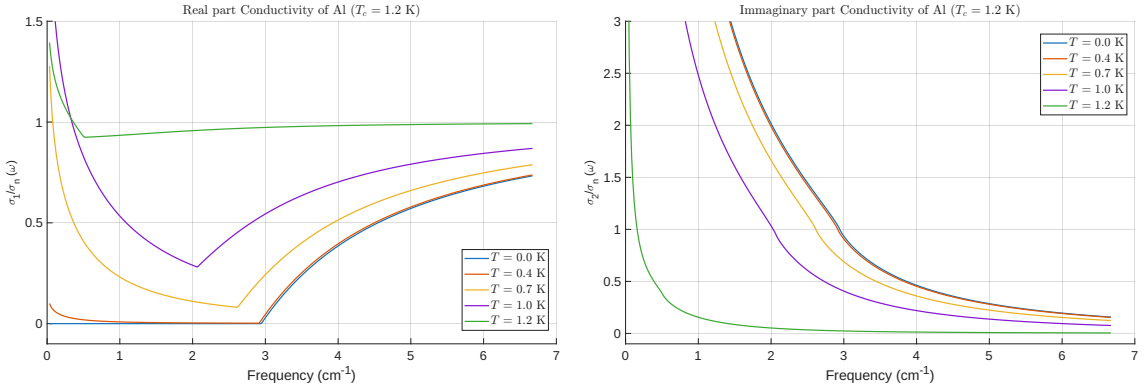


Figure 2.4: Real (sx) and imaginary (dx) conductivity for Al.

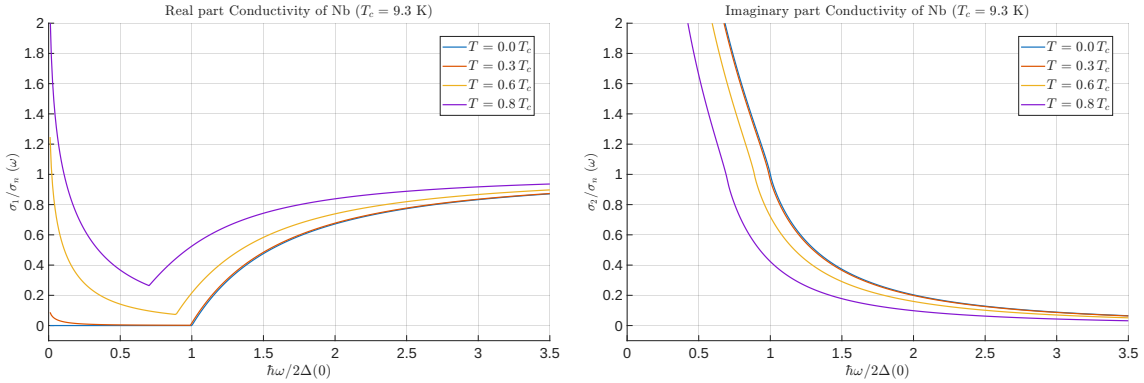


Figure 2.5: Real (sx) and imaginary (dx) conductivity for Nb in normalized units, to check the gap corresponds at $T = 0$.

To evaluate the performances of the code that i've written, i compare the numerical solutions with experimental data of Nb. To extract the data points from the figure i have used the online tool `WebPlotDigitizer`. The results are showed in the Fig.2.6 below with $T_c = 8.3$ K and $\alpha = 4.1$. We can see that the BCS understimates the value of the real conductivity σ_1, expecially for lower T, but we espected this for Niobium because is a strong-coupling superconducting material while BCS is a weak-coupling theory ($\alpha = 3.65$).

Another comparison that needs to be done is the one between the values of Δ at different temperatures extracted by the experimental data of conductivity, and the BCS mean field approximation used for the gap function in Eq. 2.2. The value of $2\Delta(T)$ is given by the frequency that corresponds to the minimum of the σ_1 at given temperature. This is showed in Fig. 2.7. Furthermore in Fig. 2.7 i have plotted the frequency at the minimum of σ_1 obtained from the numerical solution of Eq.1.1, as a double check for the code.

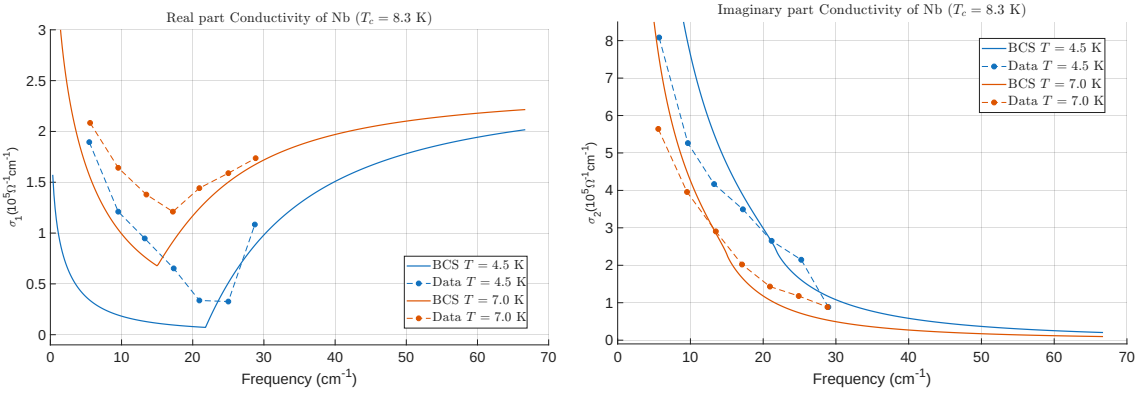


Figure 2.6: Real (sx) and imaginary (dx) conductivity for Nb compared with experimental data.

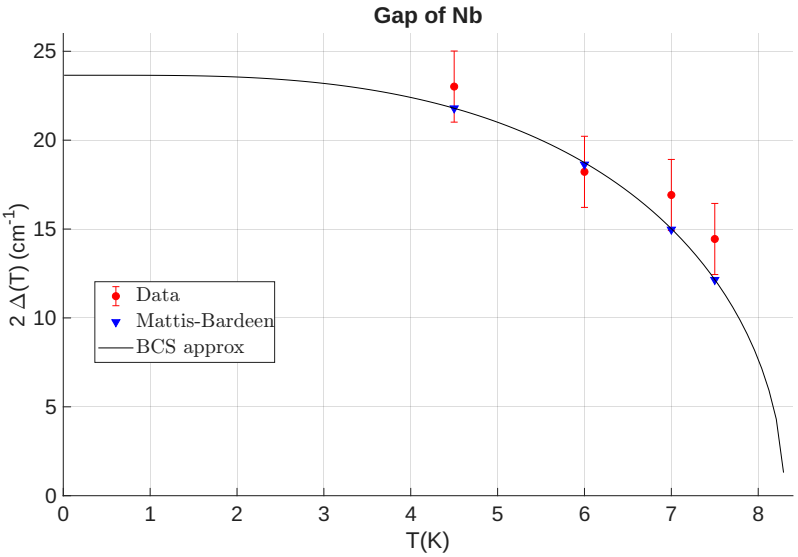


Figure 2.7: Comparison between BCS approximation of $\Delta(T)$ and the values of Δ extracted by the data. Errorbars are 2 cm^{-1} from the paper [ref].